Introduction to Virtualization

Guide to Installing Oracle VirtualBox & Ubuntu Linux

A Step-by-Step Practical Lab Guide for Students

Understanding Host vs Guest OS

The Host Operating System

- **Definition:** The primary operating system installed directly on the physical computer hardware.
- **In This Lab:** Your physical Windows computer acts as the Host.
- **Responsibility:** It manages the physical CPU, physical RAM, and storage, sharing these resources with the virtualization software.

The Guest Operating System

- **Definition:** An isolated operating system running inside a virtual machine environment.
- **In This Lab:** Ubuntu Linux will be our Guest OS.
- **Behavior:** It operates independently as if it were on a separate physical machine, completely unaware that it is virtualized.

Pre-Requisites & System Check

1. Hardware Resources Check

Ensure your Windows host has a minimum of 8 GB RAM (16 GB recommended) and at least 30 GB of free disk space.

2. Enable Hardware Virtualization

Must be active in BIOS/UEFI (Intel VT-x or AMD-V).
Verification: Open Task Manager -> Performance -> CPU -> Verify 'Virtualization: Enabled'.

3. Download Required Materials

- Oracle VirtualBox Windows Installer from official website.
- Ubuntu Desktop LTS ISO File from ubuntu.com.

Phase 1: Installing Oracle VirtualBox

Step 1: Download & Execute Installer

Navigate to virtualbox.org and download the Windows binary. Double-click the downloaded .exe installer file to begin.

Step 2: Follow Installation Wizard

Proceed with standard default options. Click 'Yes' if prompted with warnings regarding brief network adapter disconnections.

Step 3: Verification

Launch the Oracle VM VirtualBox Manager application from your desktop shortcut or start menu to confirm a successful install.

Phase 2: Provisioning the Ubuntu VM

1. Create New Instance

Click the 'New' button in the VirtualBox dashboard. Name it 'Ubuntu'.

2. Mount Operating System ISO

In the ISO Image field, browse and select the downloaded Ubuntu Desktop .iso file. Check 'Skip Unattended Installation'.

3. Allocate Base Hardware

- Memory: Assign a minimum of 4096 MB (4 GB RAM).
- Processors: Allocate at least 2 CPU Cores to ensure baseline OS stability.

4. Define Virtual Hard Storage

Select 'Create a Virtual Hard Disk Now' and allocate a minimum capacity of 25 GB.

Phase 3: Installing the Guest OS (Ubuntu)

Step 1: Boot the Virtual Environment

Select your configured Ubuntu instance from the sidebar and click the green 'Start' arrow.

Step 2: Choose Boot Environment

Use the keyboard arrow keys to select 'Try or Install Ubuntu' and hit Enter.

Step 3: Internal Setup Wizard

Follow the guided on-screen graphical installation. Select Language preferences -> choose 'Normal Installation' -> click 'Install Now'.

Step 4: Finalize Setup

Provide custom username/password credentials when prompted. Allow the files to copy, then select 'Restart Now' inside the VM box when completed.

Common Troubleshooting Tips

Issue: VM crashes or 64-bit options missing

Fix: Virtualization is disabled in the BIOS. Reboot your host computer, enter BIOS/UEFI settings (F2/F10/Del), find VT-x/AMD-V/SVM mode, and set to Enabled.

Issue: Host system lags severely when VM runs

Fix: Over-allocation of resources. Shut down the VM, open its settings, and reduce the RAM or CPU core count to stay out of the red slider zones.

Issue: Display screen size is too small / Mouse lagging

Fix: Go to the VM menu -> Devices -> Click 'Insert Guest Additions CD Image'. Run the installer within Ubuntu to install integration drivers.